Florian Tambon

701-50 Avenue Willowdale, H3T1G1, Montréal, Canada | tambonflorian01@gmail.com

+1 438-346-7594 | https://yourwebsite.com/ | | |

Education

PhD Software Engineering, Polytechnique Montréal - Canada Sept 2020 – Sept 2024 (Expected)

Supervisors: Pr. Foutse Khomh and Pr. Giuliano Antoniol

Thesis: **Who Tests the Testers? Assessing the Effectiveness and Trustworthiness of Deep Learning Model Testing Techniques**. Proposing frameworks to improve the trustworthiness and effectiveness of testing techniques applied to deep learning models on the topics of stochasticity, cost of testing or evaluation metric fine-graininess.

- Mutation testing applied to the case of deep reinforcement learning
- Accounting for stochasticity of mutation testing in supervised learning through a Bayesian formulation
- Evaluating difficulty of programming tasks within benchmarks for large language models
- Reducing the cost of testing techniques by leveraging test transfers across models

Master of Engineering, KyuTech - Japan Sept 2018 – March 2020

Double degree - Specialization in Machine Learning

Engineering Degree, Ecoles des Mines de Saint-Etienne - France Sept 2016 – March 2020

General courses in mathematics, physics, computer science, corporate management and communication tools. Specialization in Information Technology and Artificial Intelligence.

Preparatory Classes, Lycee Thiers - France Sept 2014 – March 2016

Preparation for nationwide competitive entrance exams to engineering schools. Fundamental courses in mathematics, physics and engineering.

Research Projects

Part of my PhD funding was granted by the DEpendable & Explainable Learning (DEEL) project. This was the opportunity for several collaborations with academics and aerospace industrial partners.

Non-invasive Vigilance Decline Detection Sept 2023 – Present

- Simulating a monotonous driving environment
- Acquire diverse biosignals (ECG, EEG, Eye tracking, Temperature etc.)
- Relate biosignals with vigilance declined state
- Link invasive signals (e.g, EEG, ECG) with non-invasive ones (e.g., Eye tracking, Temperature) to rely only on non-invasive signals during operations

Predicting Lightning Strike On Airplane Components Sept 2022 – Sept 2023

- Analyze historical lightning strike data on airplanes to extract and process relevant features
- Formulate the task as a machine-learning one-class anomaly detection problem
- Use the Local Outlier Factor (anomaly detection) model to decide if new airplane parts are outliers or not
- Complement the model with the SHAP approach to provide explainable details on the prediction of the model

Research Intern, Polytechnique Montréal – Canada June 2018 – Sept 2018

Technical debt identification in source code using neural networks

Teaching Experience

Teaching Assistant, Polytechnique Montréal – Canada Autumn 2022, Autumn 2023

Course: **Introduction to Programming Language**

- Conceive, teach and grade practical labs for the undergraduate course
- Basis of programming using Python. Program structures (if/else, loops etc.), Algorithms, Scientific Libraries and Basics of Oriented Objects Programming.

Teaching Assistant, Polytechnique Montréal – Canada Autumn 2021

Course: **Methods for Testing and Validating Software**

- Conceive, teach and grade practical labs for the undergraduate course
- Coverage Testing, Control Flow Graph, Unit Testing/Mock Testing, Oriented Object Testing, Logic Testing etc.

Selected Publications

Tambon, F., Laberge, G., An, L., Nikanjam, A., Mindom, P. S. N., Pequignot, Y., ... & Laviolette, F. (2022). *How to certify machine learning based safety-critical systems? A systematic literature review*. Automated Software Engineering, 29(2), 38.

Tambon, F., Majdinasab, V., Nikanjam, A., Khomh, F., & Antoniol, G. (2023, April). *Mutation testing of deep reinforcement learning based on real faults*. In 2023 IEEE Conference on Software Testing, Verification and Validation (ICST) (pp. 188-198). IEEE.

Tambon, F., Khomh, F., & Antoniol, G. (2023). *GIST: Generated Inputs Sets Transferability in Deep Learning*. ACM Transactions on Software Engineering and Methodology.

Tambon, F., Nikanjam, A., An, L., Khomh, F., & Antoniol, G. (2024). *Silent bugs in deep learning frameworks: an empirical study of keras and tensorflow*. Empirical Software Engineering, 29(1)

Tambon, F., Dakhel, A. M., Nikanjam, A., Khomh, F., Desmarais, M. C., & Antoniol, G. (2024). *Bugs in Large Language Models generated code.* [Submitted for review, Empirical Software Engineering journal]

Mahu, A., Singh, A., Tambon, F., Ouellette, B., Delisle, J. F., Paul, T. S., ... & Doyon-Poulin, P. (2024). *Validation of Vigilance Decline Capability in a Simulated Test Environment: A Preliminary Step Towards Neuroadaptive Control*. Neuroergonomics and Cognitive Engineering, 45.

Skills

THIS IS A TEST THIS IS A TEST. THIS IS A TEST. THIS IS A TEST

Referees

Pr. Foutse Khomh, Full Professor of Software Engineering, Polytechnique Montréal. email: foutse.khomh@polymtl.ca

Pr. Giuliano Antonio, (Retired) Full Professor of Software Engineering, Polytechnique Montréal. email: giuliano.antoniol@polymtl.ca

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